GlucoTwin

System Architecture

A Continuously Updating, Patient-Conditioned Digital Twin for Glucose Event Prediction

Research Prototype
Synthetic Data Only
Not a medical device

Input Data

1. Historical Profile
(Synthetic EHR-like Data)

- Age
- Sex Category
- Diabetes Duration
- Fasting Glucose
- HbA1c
- Baseline Statistics
- Diagnosis (fictional)
- Medication Category (fictional)

2. Sensor Stream
(Synthetic CGM + Activity)

- Glucose (5-min interval)
- Heart Rate
- Step Count
- Meal Annotations
- Missing Values
- Sensor Gap (per patient)
- 7 Days per Patient
- 160 Patients
- 322,560 Total Rows

Streamlit Application (Front End)



Real-Time Inference Pipeline (Per Timestamp)

Data Loader & Replay State

- Load selected patient
- Maintain replay cursor
- Provide observed prefix
- Handle missing values
- Session state per patient

Data-Quality Gate

- Check recent missingness
- Check time since last actual reading
- Withhold prediction if thresholds exceeded
- Provide quality status

Feature Engineering (22 Features)

- Recent level, mean, variability 15-min slope
- Activity and heart rate
- Meal timing
- Historical patient features
- Baseline-normalized values
- Missingness indicators
- Use only past/current data (no future leakage)

Trained Model (Scikit-learn HGB)

- Histogram Gradient Boosting Classifier
- Patient-conditioned features
- Sigmoid calibration
- Predict probability of glucose > 180 mg/dL within next 120 minutes

Prediction Output

- **Event Probability** (e.g., 28%)
- **Forecast Horizon** (120 minutes)
- **Review Category** (Normal / Review / High)
- **Data-Quality Status**
- **Latest Observation** (e.g., 142 mg/dL)

Output & Storage

SQLite Audit Log

- Patient ID
- Timestamp
- Observed Glucose
- Event Estimate
- Horizon
- Review Category
- Model Version
- **No duplicate entries** (patient + time + model)
- CSV export

Retrospective View (Ground Truth)

- Full synthetic timeline
- Visible only after explicit user action
- For review and analysis (not used for prediction)

Training and Evaluation Pipeline (Offline)

Synthetic Data Generator

- 160 patients, 7 days
- 5-minute interval
- Meal-response model
- Activity, circadian rhythm
- Missing values and pensor gaps
- Fixed random seed (2026)

Feature & Target Construction

- Causal rolling features
- Future-event label (> 180 mg/dL in next 120 min)
- Exclude current reading from label
- 15-minute training anchors after 60-minute warmup

Patient-Level Partitioning

- 88 Training Patients
- 24 Validation Patients
- 24 Calibration Patients
- 24 Test Patients

Model Training

- Scikit-learn HGB classifier
- Select leaf count using validation AP
- Sigmoid calibration on separate patients
- Rule baseline
- Dynamic-only ablation

Evaluation

- Precision, Recall, F1
- Average Precision (PR-AUC)
- Brier Score and Calibration
- Confusion Matrix
- Below-threshold subset (early warning)
- Feature Importance (permutation)
- Save metrics.json and test predictions

Artifacts

- model.joblib (estimator + calibration)
- metrics.json (evaluation results)
- test_predictions.csv (held-out outcomes)
- patients.csv (historical profiles)
- sensors/*.csv.gz (per-patient timelines)